Grade 10 Term 1 Practice Activities Summary
Decision Analysis Under Uncertainty

Contents

1 G10_T1_L1_C1C2_practice_activity1

This activity develops C1–C2 by moving from the basic language of decisions under uncertainty to complete payoff-table construction. Students identify alternatives, states of nature, uncertain events, consequences, and economic or financial context, then organise supplied or calculated payoffs in tables. The progression begins with general symbolic problems, continues through directly supplied payoffs, and then separates alternative-based and state-based variable and fixed payoff components before combining them in increasingly complete payoff structures.

1. **Introduction.**
   (a) **Introductory General Problem (C1–C2):** Students interpret a generic two-alternative, two-state decision, identify the decision elements, and place four directly supplied payoffs in the correct rows and columns of a payoff table.
   (b) **Advanced General Problem (C1–C2):** Students interpret a generic two-alternative, two-state decision with alternative–state quantities, alternative-based and state-based variable values, and alternative-based and state-based fixed values, then calculate and organise each payoff using the full component formula.
2. **Direct Alternative–State Payoffs.**
   (a) **Local Café Investment (C1–C2):** Students identify the café menu alternatives, foot-traffic states, consequences, and directly stated profit payoffs, then build the corresponding payoff table.
   (b) **Green Energy Production Mix (C1–C2):** Students compare production-mix alternatives across demand states by classifying the decision elements and arranging directly supplied payoffs.
   (c) **Global Shipping Network Design (C1–C2):** Students organise shipping-network alternatives across fuel-and-delay states and place the supplied profit outcomes in a payoff table.

3. **Alternative-Based Variable Payoff: Profit per Item Times Number of Items.**
   (a) **Community Bakery Lunch Boxes (C1–C2):** Students identify lunch-box alternatives and demand states, calculate payoffs from alternative-specific profit per item times quantities, and organise them in a table.
   (b) **Rural Internet Subscriptions (C1–C2):** Students classify subscription alternatives and adoption states, then use alternative-based per-subscriber payoffs with state-dependent subscriber counts to complete the payoff table.
4. **State-Based Variable Payoff: Profit per Item Times Number of Items.**
   (a) **Coffee Export Routes (C1–C2):** Students analyse route alternatives and market states, applying state-based per-unit values to quantities in each alternative–state pair.
   (b) **Lake Ferry Tourism (C1–C2):** Students interpret ferry-service alternatives and tourism states, then calculate and tabulate payoffs using state-based variable rates.
5. **Alternative-Based Fixed Payoff Only.**
   (a) **School Meal Service Contracts (C1–C2):** Students identify service alternatives and demand states where the payoff is fixed by the selected alternative, then place those fixed values across the states.
   (b) **Mobile Clinic Service Contracts (C1–C2):** Students compare clinic-service contract alternatives and uncertain states using alternative-based fixed payoff values.
6. **State-Based Fixed Payoff Only.**
   (a) **Neighborhood Shelter Support (C1–C2):** Students identify support alternatives and need states where payoffs are fixed by the state, then organise the repeated state-based values by column.
   (b) **Community Hall Emergency Rentals (C1–C2):** Students classify rental alternatives and emergency states and construct a payoff table from state-based fixed payoffs.
7. **Alternative-Based Variable Payoff Plus State-Based Variable Payoff.**
   (a) **Recycling Collection Contract (C1–C2):** Students combine alternative-based and state-based variable rates with quantities to calculate each contract payoff.
   (b) **Museum Ticket Platform (C1–C2):** Students use both platform-specific and state-specific per-ticket components to compute and organise ticket-platform payoffs.
8. **Alternative-Based Variable Payoff Plus Alternative-Based Fixed Payoff.**
   (a) **Local Printing Contracts (C1–C2):** Students combine alternative-based variable payoff per unit with alternative-based fixed payoff components for each demand state.
   (b) **Cold-Storage Contracts (C1–C2):** Students calculate payoffs for storage alternatives by adding alternative-specific fixed terms to alternative-based variable components.
9. **Alternative-Based Variable Payoff Plus State-Based Fixed Payoff.**
   (a) **Urban Landscaping Bids (C1–C2):** Students calculate bid payoffs using alternative-based variable components and fixed values determined by the realised state.

(b) **Solar Installation Bids (C1–C2):** Students combine installation alternative variable payoffs with state-based fixed terms and organise the results.

10. **State-Based Variable Payoff Plus Alternative-Based Fixed Payoff.**

    (a) **Regional Bus Charters (C1–C2):** Students calculate charter payoffs from state-based variable rates plus alternative-based fixed amounts.

    (b) **Air Cargo Charters (C1–C2):** Students interpret cargo alternatives and states, then combine state-based variable and alternative-based fixed components.

11. **State-Based Variable Payoff Plus State-Based Fixed Payoff.**

    (a) **Fruit Export Permits (C1–C2):** Students calculate payoffs using variable and fixed components that both depend on the state of nature.

    (b) **Seafood Export Licenses (C1–C2):** Students build payoff tables by applying state-based variable and state-based fixed values across export-license alternatives.

12. **Alternative-Based Fixed Payoff Plus State-Based Fixed Payoff.**

    (a) **Warehouse Security Agreements (C1–C2):** Students add alternative-based fixed and state-based fixed components for each security-agreement outcome.

    (b) **Harbor Inspection Agreements (C1–C2):** Students classify inspection alternatives and states, then combine the two fixed payoff components in a table.

13. **Alternative-Based Variable Payoff Plus State-Based Variable Payoff Plus Alternative-Based Fixed Payoff.**

    (a) **Water Treatment Modules (C1–C2):** Students combine quantities with both variable rates and add an alternative-based fixed component.

    (b) **Freight Locker Network (C1–C2):** Students calculate network payoffs from alternative-based variable, state-based variable, and alternative-based fixed components.

14. **Alternative-Based Variable Payoff Plus State-Based Variable Payoff Plus State-Based Fixed Payoff.**

    (a) **District Heating Supply (C1–C2):** Students combine both variable payoff components with a fixed component determined by the state.

    (b) **Port Cold-Chain Service (C1–C2):** Students compute service payoffs using alternative-based and state-based variable terms plus state-based fixed values.

15. **Alternative-Based Variable Payoff Plus Alternative-Based Fixed Payoff Plus State-Based Fixed Payoff.**

    (a) **Pharmacy Delivery Plan (C1–C2):** Students calculate delivery-plan payoffs from an alternative-based variable rate, an alternative fixed term, and a state fixed term.

    (b) **Timber Processing Line (C1–C2):** Students organise payoffs for processing-line alternatives by combining the same three component types.

16. **State-Based Variable Payoff Plus Alternative-Based Fixed Payoff Plus State-Based Fixed Payoff.**

(a) **Public Library Outreach (C1–C2):** Students compute outreach payoffs from state-based variable rates and both alternative-based and state-based fixed components.

(b) **Grain Terminal Schedule (C1–C2):** Students calculate and tabulate terminal-schedule payoffs with state-based variable, alternative fixed, and state fixed components.

17. **Alternative-Based and State-Based Variable Payoffs Plus Alternative-Based and State-Based Fixed Payoffs.**

(a) **Urban Food Hub (C1–C2):** Students use the complete payoff structure with quantities, both variable components, and both fixed components to build the table.

(b) **Regional Rail Cargo (C1–C2):** Students finish the progression by calculating rail-cargo payoffs from all four payoff-component types and arranging the final comparison table.

2 G10_T1_L1_C1C3_practice_activity2

This activity develops C1 and C3 by asking students to identify decision elements and calculate profit payoffs from revenue, fixed-cost, and variable-cost information. The organisation mirrors the payoff-component taxonomy from Activity 1, but each problem requires arithmetic such as revenue minus total cost, units times unit margin, or combinations of state-based and alternative-based cost components before the final payoff table is completed.

1. **Direct Alternative–State Payoffs.**

(a) **Weekend Market Stall (C1–C3):** Two alternatives and two market states; students identify the stall choice, market activity state, and calculate each payoff as sales revenue minus total operating cost.

(b) **Youth Sports Camp (C1–C3):** Three alternatives and three enrollment states; students subtract total delivery cost from fee income for each camp and state.

2. **Alternative-Based Variable Payoffs.**

(a) **Reusable Bottle Workshop (C1–C3):** Two alternatives and two order states; students multiply units sold by the alternative-specific margin between selling price and variable cost per unit.

(b) **Community Bakery Packs (C1–C3):** Three alternatives and three demand states; students calculate pack profits from quantities times alternative-specific per-pack margins.

3. **State-Based Variable Payoffs.**

(a) **School Notebook Printing (C1–C3):** Two alternatives and two demand states; students apply state-based variable cost rates to units and subtract from revenue.

(b) **Local Juice Production (C1–C3):** Three alternatives and three demand states; students compute bottle profits using state-dependent variable costs.

4. **State-Based Fixed Payoffs.**

(a) **Museum Evening Program (C1–C3):** Two alternatives and two attendance states; students subtract state-based fixed operating costs from revenue.

(b) **Town Exhibition Plan (C1–C3):** Three alternatives and three attendance states; students calculate payoffs with fixed costs determined by state.

5. **Alternative-Based Fixed Payoffs.**
   (a) **Library Delivery Service (C1–C3):** Two alternatives and two demand states; students subtract alternative-specific fixed setup costs from revenue.
   (b) **Student Media Studio (C1–C3):** Three alternatives and three demand states; students calculate payoffs using alternative-based fixed costs.
6. **State-Based Fixed Payoff Plus Alternative-Based Fixed Payoff.**
   (a) **Community Fair Venue (C1–C3):** Two alternatives and two weather states; students subtract both state-based and alternative-based fixed costs.
   (b) **Regional Debate Tournament (C1–C3):** Three alternatives and three attendance states; students combine both fixed-cost categories before tabulating profits.
7. **Alternative-Based Variable Payoff Plus Alternative-Based Fixed Payoff.**
   (a) **Custom T-Shirt Order (C1–C3):** Two alternatives and two order states; students calculate unit margins and subtract alternative-specific fixed costs.
   (b) **Learning Kit Supplier (C1–C3):** Three alternatives and three order states; students combine alternative-based unit costs with alternative-based fixed costs.
8. **Alternative-Based Variable Payoff Plus State-Based Fixed Payoff.**
   (a) **Neighborhood Garden Boxes (C1–C3):** Two alternatives and two demand states; students combine alternative-based variable costs with fixed costs determined by state.
   (b) **School Furniture Bid (C1–C3):** Three alternatives and three demand states; students calculate payoffs from alternative variable margins and state-based fixed charges.
9. **State-Based Variable Payoff Plus Alternative-Based Fixed Payoff.**
   (a) **Festival Snack Kiosk (C1–C3):** Two alternatives and two crowd states; students apply state-based variable costs and subtract alternative fixed costs.
   (b) **After-School Transport (C1–C3):** Three alternatives and three ridership states; students combine state-based variable costs with alternative-based fixed costs.
10. **State-Based Variable Payoff Plus State-Based Fixed Payoff.**
   (a) **Farmers Market Delivery (C1–C3):** Two alternatives and two market states; students subtract state-based variable and fixed costs from revenue.
   (b) **Coastal Recycling Collection (C1–C3):** Three alternatives and three collection states; students complete payoff calculations with both cost components determined by state.

3 G10_T1_L2_C4C5_practice_activity3

This activity develops C4–C5 decision criteria for decisions without using probabilities in the selection rules, even though the payoff tables display probabilities. Students apply the optimistic Maximax rule, the conservative Maximin rule, and the Minimax Regret rule. The activity is organised as three solved decision-analysis problems, each requiring identification of best possible payoffs, worst-case payoffs, state-by-state opportunity losses, maximum regrets, and the resulting selected alternative.

1. **Problems.**

    (a) **Community Theater Ticketing Strategy Under Demand Uncertainty:** Three strategies are compared across strong and weak demand; students select by Maximax and Maximin, construct a regret table from the best payoff in each state, and choose the strategy with the smallest maximum regret.

    (b) **Holiday Inventory Plan Selection Under Demand Uncertainty:** Three inventory plans are compared across strong, moderate, and weak demand; students find optimistic and conservative choices and use opportunity losses to identify the Minimax Regret plan.

    (c) **Routing System Choice Under Fuel and Congestion Uncertainty:** Two routing systems are compared across four fuel-and-congestion states; students determine Maximax and Maximin choices, calculate state-by-state regrets, and select by smallest maximum regret.

4 G10_T1_L2_C6C7_practice_activity4

This activity develops C6–C7 by using probabilities in decision making. The first section applies Expected Monetary Value and the Bayes decision criterion with stated probabilities. The second section treats a two-state probability as a variable, builds expected-value expressions in terms of p, and uses sensitivity analysis to determine probability thresholds, preferred alternatives, indifference points, and dominated options.

1. **C6: Decision-Making Under Risk Using Probabilities.**

    (a) **Community Theater Ticketing Strategy Under Demand Uncertainty:** Three strategies are evaluated across strong and weak demand with stated probabilities; students compute expected values and choose the highest expected payoff.

    (b) **Holiday Inventory Plan Selection Under Demand Uncertainty:** Three inventory plans are evaluated across strong, moderate, and weak demand with stated probabilities; students calculate each expected value and select by the Bayes criterion.

    (c) **Routing System Choice Under Fuel and Congestion Uncertainty:** Two routing systems are evaluated across four states with stated probabilities; students compute expected values, including a negative payoff state, and choose the better system.

2. **C7: Sensitivity Analysis of Probability-Based Decisions.**

    (a) **Expected Value Comparison of Marketing Options:** Two marketing options are compared across states S_1 and S_2 with probabilities p and $1 - p$; students form expected-value expressions and identify the threshold where the preferred option changes.

    (b) **Expected Value Comparison of Production Plans:** Three production plans are compared using expected values as functions of p; students determine which plan is optimal for different probability ranges and recognise the never-optimal plan.

    (c) **Expected Value Comparison of Zoning Options:** Two zoning options are compared with variable probability p; students solve for the indifference point and state which option is preferred above or below it.

3. **Minimal Solutions.**

    (a) **Community Theater Ticketing Strategy Under Demand Uncertainty:** Provides expected-value calculations and the Strategy A choice.

(b) **Holiday Inventory Plan Selection Under Demand Uncertainty:** Provides expected-value calculations and the Plan B choice.

(c) **Routing System Choice Under Fuel and Congestion Uncertainty:** Provides expected-value calculations and the System B choice.

(d) **Expected Value Comparison of Marketing Options:** Gives expected-value formulas, the $p = 0.4$ indifference point, and the preference regions.

(e) **Expected Value Comparison of Production Plans:** Gives expected-value formulas, the $p = \frac{5}{11}$ tie between A and B, preference regions, and notes that C is never optimal.

(f) **Expected Value Comparison of Zoning Options:** Gives expected-value formulas, the $p = 0.5$ indifference point, and the preference regions.